

Flocking - Example solution

The example solution for the flocking exercises follow the *Sense-Think-Act* pattern. In the *Sense* step, the robot computes vectors of attraction and repulsion, based on the different sensor inputs. In the *Think* step, the robot computes an aggregate vector according to the needs of the mission. In the *Act* step, the robot sets the velocities of the wheels to move into the direction of the computed vector.

Hexagonal pattern formation

For this experiment, the control software is defined in the file `pattern_formation_hexagon.lua`. The robots are simply computing the Lennard-Jones potential with respect to their neighbors and are following the vector of the potential. Due to the nature of the Lennard-Jones potential the resulting grid is approximately hexagonal.

Circular pattern formation

For this experiment, the control software is defined in the file `pattern_formation_circle.lua`. The robots are computing the Lennard-Jones potential towards their neighbors and a second potential with regard to the LED. The final vector of attraction is then the sum of the two potential vectors.

Flocking

For this experiment, the control software is defined in the file `flocking.lua`. The *Think* part of the pattern is structured as a finite-state machine. The robot has two states, flocking and grouping. In grouping, the robot only considers the vector generated by the Lennard-Jones potential towards its neighbors. This results in the robot moving into the direction of its peers, while simultaneously performing obstacle avoidance. Once the robot perceived a sufficient number of neighbors for a sufficient number of consecutive time steps, it decides to switch from the grouping behavior to the flocking behavior. In the flocking behavior, the robot considers not only the vector of the Lennard-Jones potential but also the vector pointing towards the light source.